

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: **Sodium tetraborate decahydrate**  
 Product Description: **Sodium tetraborate decahydrate**  
 Cat No. : B80  
 Synonyms Sodium borate decahydrate; Borax  
 CAS No 1303-96-4  
 Molecular Formula B<sub>4</sub>Na<sub>2</sub>O<sub>7</sub>·10H<sub>2</sub>O

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company** Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**E-mail address** Enquiry.my@thermofisher.com

**Emergency Telephone Number** Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                       |                      |
|-----------------------|----------------------|
| Reproductive Toxicity | Category 1B (H360FD) |
|-----------------------|----------------------|

**Label Elements**

**Signal Word**
**Danger**
**Hazard Statements**

H360FD - May damage fertility. May damage the unborn child

# SAFETY DATA SHEET

Sodium tetraborate decahydrate

Revision Date 23-Mar-2025

## Precautionary Statements

### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P270 - Do not eat, drink or smoke when using this product  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell  
P308 + P313 - IF exposed or concerned: Get medical advice/attention

### Storage

P403 - Store in a well-ventilated place

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors  
Toxic to terrestrial vertebrates

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                 | CAS No    | Weight % |
|-------------------------------------------|-----------|----------|
| Borates, tetra, sodium salts, decahydrate | 1303-96-4 | 100      |
| Borates, tetra, sodium salts, anhydrous   | 1330-43-4 | -        |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

# SAFETY DATA SHEET

Sodium tetraborate decahydrate

Revision Date 23-Mar-2025

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Keep product and empty container away from heat and sources of ignition. Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

Oxides of boron, Sodium oxides.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### Environmental precautions

Should not be released into the environment.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

# SAFETY DATA SHEET

Sodium tetraborate decahydrate

Revision Date 23-Mar-2025

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                                 | Malaysia | ACGIH TLV                                             | OSHA PEL                            |
|-------------------------------------------|----------|-------------------------------------------------------|-------------------------------------|
| Borates, tetra, sodium salts, decahydrate |          | TWA: 2 mg/m <sup>3</sup><br>STEL: 6 mg/m <sup>3</sup> | (Vacated) TWA: 10 mg/m <sup>3</sup> |
| Borates, tetra, sodium salts, anhydrous   |          | TWA: 2 mg/m <sup>3</sup><br>STEL: 6 mg/m <sup>3</sup> | (Vacated) TWA: 10 mg/m <sup>3</sup> |

| Component                                 | European Union | The United Kingdom                                                 | Germany |
|-------------------------------------------|----------------|--------------------------------------------------------------------|---------|
| Borates, tetra, sodium salts, decahydrate |                | STEL: 15 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr |         |
| Borates, tetra, sodium salts, anhydrous   |                | STEL: 3 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr  |         |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### Recommended Filter type:

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### Appearance

White

#### Physical State

Powder Solid

#### Odor

Odorless

#### Odor Threshold

No data available

# SAFETY DATA SHEET

Sodium tetraborate decahydrate

Revision Date 23-Mar-2025

|                                           |                                                                   |                                   |
|-------------------------------------------|-------------------------------------------------------------------|-----------------------------------|
| pH                                        | 9                                                                 | 5% aq.sol. 20°C                   |
| Melting Point/Range                       | > 1000 °C / > 1832 °F                                             |                                   |
| Softening Point                           | No data available                                                 |                                   |
| Boiling Point/Range                       | No information available                                          |                                   |
| Flash Point                               | No information available                                          | Method - No information available |
| Evaporation Rate                          | Not applicable                                                    | Solid                             |
| Flammability (solid,gas)                  | No information available                                          |                                   |
| Explosion Limits                          | No data available                                                 |                                   |
| Vapor Pressure                            | No information available                                          |                                   |
| Vapor Density                             | Not applicable                                                    | Solid                             |
| Specific Gravity / Density                |                                                                   |                                   |
| Bulk Density                              | No data available                                                 |                                   |
| Water Solubility                          | 49.74 g/L (20°C)                                                  |                                   |
| Solubility in other solvents              | No information available                                          |                                   |
| Partition Coefficient (n-octanol/water)   |                                                                   |                                   |
| Component                                 | log Pow                                                           |                                   |
| Borates, tetra, sodium salts, decahydrate | - 0.757                                                           |                                   |
| Borates, tetra, sodium salts, anhydrous   | -0.7570                                                           |                                   |
| Autoignition Temperature                  | Not applicable                                                    |                                   |
| Decomposition Temperature                 | > 100°C                                                           |                                   |
| Viscosity                                 | Not applicable                                                    | Solid                             |
| Explosive Properties                      | Not explosive                                                     |                                   |
| Oxidizing Properties                      | Not oxidising                                                     |                                   |
| Molecular Formula                         | B <sub>4</sub> Na <sub>2</sub> O <sub>7</sub> ·10H <sub>2</sub> O |                                   |
| Molecular Weight                          | 381.36                                                            |                                   |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

#### Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.  
None under normal processing.

### Conditions to Avoid

Exposure to air. Incompatible products. Avoid dust formation.

# SAFETY DATA SHEET

Sodium tetraborate decahydrate

Revision Date 23-Mar-2025

## Incompatible Materials

Strong oxidizing agents. Strong acids. Finely powdered metals.

## Hazardous Decomposition Products

Oxides of boron. Sodium oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

| Component                                 | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation                        |
|-------------------------------------------|---------------------------|------------------------------|----------------------------------------|
| Borates, tetra, sodium salts, decahydrate | 5660 mg/kg ( Rat )        | > 2000 mg/kg ( Rabbit )      | 2.03 mg/l (Rat)                        |
| Borates, tetra, sodium salts, anhydrous   | LD50 = 2660 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rabbit ) | LC50 > 2 mg/m <sup>3</sup> ( Rat ) 4 h |

##### (b) skin corrosion/irritation;

No data available

##### (c) serious eye damage/irritation;

Test species

No data available

Observation end point

rabbit

Severe eye irritant  
fully reversible

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

| Component                                                      | Test method             | Test species | Study result        |
|----------------------------------------------------------------|-------------------------|--------------|---------------------|
| Borates, tetra, sodium salts, decahydrate<br>1303-96-4 ( 100 ) | OECD Test Guideline 406 | guinea pig   | - - non-sensitising |

##### (e) germ cell mutagenicity;

No data available

##### (f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

##### (g) reproductive toxicity;

Category 1B

| Component                                                      | Test method                                                 | Test species / Duration | Study result                                           |
|----------------------------------------------------------------|-------------------------------------------------------------|-------------------------|--------------------------------------------------------|
| Borates, tetra, sodium salts, decahydrate<br>1303-96-4 ( 100 ) | OECD Test Guideline 416<br>-----<br>OECD Test Guideline 414 | Rat                     | NOAEL = 9.6 mg/kg<br>-----<br>NOAEL =<br>17.5<br>mg/kg |

**Reproductive Effects**  
**Teratogenicity**

Experiments have shown reproductive toxicity effects on laboratory animals.  
May cause harm to the unborn child.

# SAFETY DATA SHEET

Sodium tetraborate decahydrate

Revision Date 23-Mar-2025

(h) STOT-single exposure; No data available

Test species / Sex / Route of exposure mouse / Inhalation

Effective dose NOAEL 0.186 mg/l/4h

(i) STOT-repeated exposure; No data available

Test species / Duration Rat

Study result NOAEL = 118 mg/kg

Target Organs None known.

(j) aspiration hazard; Not applicable

Solid

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed No information available.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

| Component                                 | Freshwater Fish                                                   | Water Flea                                        | Freshwater Algae                                                                                                                      | Microtox |
|-------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------|
| Borates, tetra, sodium salts, decahydrate | 340 mg/L LC50 96 h<br>708 mg/l LC50 96 h<br>(Pimephales promelas) | 1085 - 1402 mg/L LC50<br>48 h                     | 2.6-21.8 mg/L EC50 96h<br>158 mg/L EC50 = 96h                                                                                         | -        |
| Borates, tetra, sodium salts, anhydrous   | LC50: = 340 mg/L, 96h<br>(Limanda limanda)                        | LC50: 1085 - 1402<br>mg/L, 48h (Daphnia<br>magna) | EC50: 2.6 - 21.8 mg/L,<br>96h static<br>(Pseudokirchneriella<br>subcapitata)<br>EC50: = 158 mg/L, 96h<br>(Desmodesmus<br>subspicatus) |          |

### Persistence and degradability

Persistence Persistence is unlikely.

Degradability Not relevant for inorganic substances.

### Bioaccumulative potential

Bioaccumulation is unlikely

| Component                                 | log Pow | Bioconcentration factor (BCF) |
|-------------------------------------------|---------|-------------------------------|
| Borates, tetra, sodium salts, decahydrate | - 0.757 | No data available             |
| Borates, tetra, sodium salts, anhydrous   | -0.7570 | No data available             |

### Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

# SAFETY DATA SHEET

Sodium tetraborate decahydrate

Revision Date 23-Mar-2025

Other adverse effects No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

#### **Other Information**

Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

### Road and Rail Transport

Not regulated

### IATA

Not regulated

### **Special Precautions for User**

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### **International Inventories**

X = listed

| Component                                 | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-------------------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Borates, tetra, sodium salts, decahydrate | 215-540-4 | X    | X   | X     | X    | X    | X     | X    | KE-03483 |
| Borates, tetra, sodium salts, anhydrous   | 215-540-4 | X    | X   | X     | X    | X    | X     | X    | KE-12384 |

### National Regulations

#### **Persistent Organic Pollutant Ozone Depletion Potential**

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

| Component                                 | Persistent Organic Pollutant | Ozone Depletion Potential | Pesticides Act 1974 |
|-------------------------------------------|------------------------------|---------------------------|---------------------|
| Borates, tetra, sodium salts, decahydrate |                              |                           | X                   |
| Borates, tetra, sodium salts, anhydrous   |                              |                           | X                   |

## SECTION 16: OTHER INFORMATION



# SAFETY DATA SHEET

Sodium tetraborate decahydrate

Revision Date 23-Mar-2025

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

23-Mar-2025

Revision Summary

SDS sections updated.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**